

Descending locus coeruleus noradrenergic signaling to spinal astrocyte subset is required for stress-induced pain facilitation

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This **important** study identifies a novel role for Hes5+ astrocytes in modulating the activity of descending pain-inhibitory noradrenergic neurons from the locus coeruleus during stress-induced pain facilitation. The role of glia in modulating neurological circuits including pain is poorly understood, and in that light, the role of Hes5+ astrocytes in this circuit is a key finding with broader potential impacts. However, the impact of this work is limited by **incomplete** evidence, notably the fact that acute restraint stress is generally anti-nociceptive rather than pro-nociceptive, and a lack of specificity in defining this novel circuit.

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Abstract

It is known that stress powerfully alters pain, but its underlying mechanisms remain elusive. Here, we identified a circuit, locus coeruleus descending noradrenergic neurons projecting to the spinal dorsal horn (LC $\rightarrow$ SDH-NA neurons), that is activated by acute exposure to restraint stress and is required for stress-induced mechanical pain hypersensitivity in mice. Interestingly, the primary target of spinal NA released from descending LC $\rightarrow$ SDH-NAergic terminals causing the stress-induced pain hypersensitivity was α_{1A} -adrenaline receptors (α_{1A} Rs) in Hes5-positive (Hes5 $^{+}$) astrocytes located in the SDH, an astrocyte subset that has an ability to induce pain sensitization. Furthermore, activation of Hes5 $^{+}$ astrocytes reduced activity of SDH-inhibitory neurons (SDH-INs) that have an inhibitory role in pain processing. This astrocytic reduction of IN activity was canceled by an A $_1$ -adenosine receptor (A $_1$ R)-knockdown in SDH-INs, and the A $_1$ R-knockdown suppressed pain hypersensitivity caused by

acute restraint stress. Therefore, our findings suggest that LC $\rightarrow$ SDH-NA neuronal signaling to Hes5⁺ SDH astrocytes and subsequent astrocytic reduction of SDH-IN activity are essential for pain facilitation caused by stress.

Introduction

Somatosensory information generated at the periphery, such as skin, is conveyed to the dorsal horn of the spinal cord (SDH) via primary afferent sensory neurons and further to the brain^{1,2}. The SDH is the first region for processing the peripherally derived sensory information and thus is critical for its proper transmission to and perception in the brain^{1,3-6}. The processing of somatosensory information is not only tightly regulated by complex neuronal circuits within the SDH but also powerfully modulated by descending neurons from the brain^{7,8}. The locus coeruleus (LC), a small nucleus located in the brainstem, modulates pain information processing in the SDH via descending noradrenaline (NA) neurons^{8,9-11}. Activation of SDH-projecting LC-NA (LC $\rightarrow$ SDH-NA) neurons leads to a release of NA in the SDH¹². Spinal NA acts on α_1 -adrenaline receptors (α_1 Rs) expressed on inhibitory neurons (INs)^{13,14} and facilitates γ -aminobutyric acid (GABA)-mediated inhibitory synaptic transmission^{15,16}. This NA signal has been implicated in the inhibitory control of nociceptive information processing^{9,11}. Besides neurons, glial cells in the SDH, such as microglia and astrocytes, also have a remarkable ability to modulate the processing and transmission of somatosensory information^{17,18-19}. Glial modulation has been extensively studied in pathological settings^{17,19}, but studies using *in vivo* imaging have shown that SDH astrocytes respond to noxious stimuli applied to the periphery under normal conditions²⁰⁻²³. Furthermore, we have recently identified a subset of spinal astrocytes defined by expression of hairy and enhancer of split 5 (Hes5) as a new target of NA²². Astrocytes express various types of adrenaline receptors²⁴; activation of LC $\rightarrow$ SDH-NA neurons induces an increase in intracellular Ca²⁺ ([Ca²⁺]_i), an indicator of astrocyte activity²⁵ via α_1 Rs²², which are the most abundant adrenaline receptors in astrocytes^{26,27}. Hes5⁺ astrocytes are selectively distributed in the superficial laminae of SDH where it receives primary afferent sensory fibers. Activation of α_1 Rs of Hes5⁺ SDH astrocytes by intrathecal administration of NA induces pain sensitization to light mechanical stimuli applied to the skin²². Given that most of the varicosities of NAergic axons/terminals are located closer to astrocytic processes than to neuronal synapses²⁸, Hes5⁺ astrocytes could be a critical NA target for modulating pain in the SDH.

LC-NA neurons are known to respond to several external and internal stressors²⁹. In rodent models of stress, nociceptive stimuli (e.g., electric shock, animal bite, etc.) are frequently employed as stressors. LC-NA neurons respond not only to stimuli that directly excite nociceptors^{30,31}, but also to stressors that do not involve their direct stimulation^{30,32}. Several studies using rodent stress models have shown that exposure to stressors with or without nociceptor activation induces a remarkable change in pain-associated behavior^{33,34}. Despite the impact of stress on the pain system, the mechanism underlying a link between stress and pain, especially pain exacerbation, at the neural circuit and cellular levels remains poorly understood, although it has been proposed that activation of enkephalinergic and GABAergic INs in the SDH is involved in stress-induced pain inhibition³⁵. Since stressful events are known to have a significant impact on chronic pain^{36,37}, providing new insights into the mechanisms by which stress modulates pain is also of clinical importance.

In this study, we investigated how stress modulates pain with a focus on LC $\rightarrow$ SDH-NA neurons and their target Hes5⁺ astrocytes in the SDH, using multiple approaches. Our experiments using *in vivo* Ca²⁺ imaging, chemogenetics and optogenetics, cell ablation, electrophysiology, biochemical imaging for NA release and astrocytic Ca²⁺ responses, conditional gene knockout, and behavioral

analyses demonstrate for the first time that LC^{→SDH}-NA neuronal signaling to Hes5⁺ SDH astrocytes and subsequent astrocytic reduction of SDH-IN activity are essential for stress-induced mechanical pain hypersensitivity.

Results

LC^{→SDH}-NA neurons mediate mechanical hypersensitivity after acute restraint stress

To investigate the mechanism underlying stress-induced pain modulation, we subjected wild-type (WT) mice to acute restraint stress exposure (**Figure 1A**), a well-known and frequently used stress model²⁹. Consistent with a previous study³⁸, restraint stress for 30 min and 1 hour induced hypersensitivity to light mechanical stimuli (by applying von Frey filaments) to the plantar skin of the hindpaw (**Figure 1B**). However, with longer exposure (2 hours) to restraint stress, mechanical hypersensitivity was attenuated (**Figure S1**). In subsequent experiments, we chose 1 hour as the exposure time to restraint stress. To analyze LC-NA neuronal activity during restraint stress, we performed fiber photometry using *NET-Cre* mice (Cre is expressed in NAergic neurons under the control of the promoter of NA transporter [NET])³⁹ with microinjection into the LC of an adeno-associated viral (AAV) vector designed to express GCaMP6s (genetic encoded fluorescent Ca²⁺ indicator) in a Cre-dependent manner (AAV-flex[GCaMP6s]). Three weeks after the injection, GCaMP6s was expressed selectively in tyrosine hydroxylase⁺ (TH⁺) LC-NA neurons (**Figure 1C**)⁴⁰. We found that restraint stress exposure increased the number of Ca²⁺ events in these neurons (**Figure 1, D** and **E**; **Figure S2**). The number of Ca²⁺ events was highest during the first 20 min of the restraint stress exposure and gradually decreased later.

To investigate the role of LC-NA neurons in mechanical hypersensitivity, we utilized a cell ablation approach⁴¹; diphtheria toxin (DTX) receptors (DTR) were specifically expressed in LC-NA neurons by injection of AAV-flex[DTR-EGFP] into the LC of *NET-Cre* mice (**Figure 1F**). DTR⁺ neurons (detected by GFP) mainly co-expressed TH, and these neurons were ablated by DTX administration (10 µg/kg, two injections 24-hour apart), which resulted in a loss of TH signal in the LC and reduction of NET signal in the SDH (**Figure 1G**). We performed behavioral tests in these mice and observed that ablation of LC-NA neurons abolished mechanical hypersensitivity after restraint stress exposure (**Figure 1H**). To specifically ablate LC^{→SDH}-NA neurons, a retrograde AAV vector expressing Cre (AAVretro-Cre) was injected into the L4 SDH, followed by AAV-flex[DTR-EGFP] injection into the LC (**Figure 1I**). In these mice, TH⁺ LC^{→SDH}-NA neurons expressing DTR (detected by EGFP) were ablated after DTX administration (**Figure 1J**). These mice also failed to induce mechanical hypersensitivity after the restraint stress (**Figure 1K**). These results suggest that descending LC^{→SDH}-NA neurons are necessary for mechanical pain hypersensitivity after acute restraint stress exposure. Furthermore, we tested whether optogenetic activation of LC^{→SDH}-NA neurons could induce mechanical hypersensitivity, using ChrimsonR, a well-established red-shifted channelrhodopsin⁴². *NET-Cre* mice were microinjected with AAV-flex[ChrimsonR-tdTomato] into the LC (**Figure 1L**). Three weeks later, we confirmed expression of ChrimsonR (detected by tdTomato) in TH⁺ neurons in the LC and NET⁺ axons/terminals in the SDH (**Figure 1M**). Optogenetic stimulation of LC^{→SDH}-NA neuronal axons/terminals in the SDH evoked mechanical hypersensitivity in these mice (**Figure 1N**). These results together demonstrate that LC^{→SDH}-NA neurons are activated by acute stress exposure and are not only necessary for stress-induced mechanical pain hypersensitivity but also sufficient to phenocopy the behavioral response.

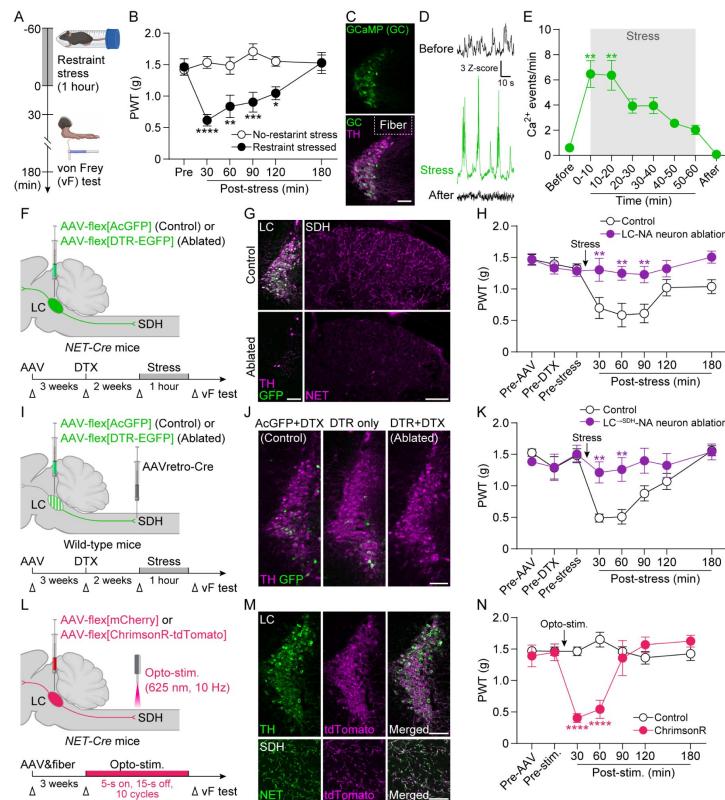


Figure 1.

LC-SDH-NA neurons mediate stress-induced mechanical hypersensitivity

(A) Schematic illustration of an experiment to investigate the effects of acute exposure to restraint stress (1 hour) on mechanosensory behavior in mice, using von Frey (vF) filaments. (B) Change in paw withdrawal threshold (PWT) measured by vF filaments in wild-type mice after restraint stress ($n = 6$ mice per group; two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$ vs. no-restraint stress group). (C) Expression of GCaMP6s (GC; green) in the LC at 3 weeks after intra-LC injection of AAV-flex[GCaMP6s] in *NET-Cre* mice. TH immunofluorescence is shown in magenta. Dashed line indicates the location of the implanted optic fiber. Scale bar, 100 μ m. (D and E) Representative traces and change in the frequency of GCaMP6s signals in LC-NA neurons ($n = 6$ mice; Friedman test with post hoc Dunn's multiple comparisons test; $**P < 0.01$ vs. the data of 'Before'). Traces shown at the top, middle, and bottom (D) indicate Ca^{2+} signals before, during, and after restraint stress, respectively. (F) Schematic illustration of the strategy of ablating LC-NA neurons using AAV vectors incorporating DTR (fused with EGFP) injected into the LC in *NET-Cre* mice. (G) TH immunofluorescence (magenta) and GFP (green) in the LC (left) and NET immunofluorescence (magenta) in the SDH (right) after administration of DTX (10 μ g/kg, i.p., two injections 24 h-apart) in control mice (top) and DTR-expressing mice (bottom). Scale bars, 100 μ m. (H) Effects of ablation of LC-NA neurons on PWT changes after acute restraint stress ($n = 7$ mice per group; two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $**P < 0.01$ vs. control group). (I) Schematic illustration of the strategy of ablating LC-SDH-NA neurons using a retrograde AAV vector incorporating Cre injected into the SDH and an AAV vector incorporating DTR (fused with EGFP) injected into the LC in wild-type mice. (J) Representative images of LC-SDH-NA neurons in control or DTR-expressing mice treated with vehicle or DTX administration, respectively. GFP (green) and TH (magenta). Scale bar, 100 μ m. (K) Effect of ablation of LC-SDH-NA neurons on PWT changes after restraint stress ($n = 11$ mice per group; two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $**P < 0.01$ vs. control group). (L) Schematic illustration of the strategy for activating LC-SDH-NA neuronal axons/terminals using an AAV vector incorporating ChrimsonR (fused with tdTomato) injected into the LC in *NET-Cre* mice and of an optic cannula implanted in the SDH. (M) Representative images of TH (green) and tdTomato (magenta) expression in the LC (top) and NET (green) and tdTomato (magenta) expression in the SDH (bottom) at 3 weeks after intra-LC injection of AAV-flex[ChrimsonR-tdTomato] in *NET-Cre* mice. Scale bars, 100 μ m (top) and 50 μ m (bottom). (N) PWT before and after optogenetic stimulation (opto-stim.) in LC-SDH-NA axons/terminals (625 nm, 2 mW, 10 Hz, 5-ms pulse duration, 5-s light on, 15-s light off, 10 cycles) (Control, $n = 4$ mice; ChrimsonR, $n = 5$ mice; two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $****P < 0.0001$ vs. control group). Data represent mean $\pm$ SEM. This figure was created using [BioRender.com](https://www.biorender.com). See also [Figure S1](#) and [S2](#).

LC $\rightarrow$ SDH-NAergic signaling to Hes5 $^{+}$ SDH astrocytes is required for stress-induced mechanical hypersensitivity

To investigate the mechanism by which activation of LC $\rightarrow$ SDH-NA neurons induces mechanical hypersensitivity, we first examined if NA is released in the SDH when these neurons are stimulated, using the genetically encoded fluorescent NA sensor GRAB_{NE1m}⁴³. In spinal cord slices from *NET-Cre* mice that had been injected AAV-flex[ChrimsonR-tdTomato] into the LC and AAV-gfaABC₁D-GRAB_{NE1m} (expressing GRAB_{NE1m} under the control of the astrocyte-selective promoter gfaABC₁D⁴⁴ into the SDH) (**Figure 2A**), optogenetic stimulation of LC $\rightarrow$ SDH-NA axons/terminals increased GRAB_{NE1m} fluorescence intensity (**Figure 2B**), confirming NA release from LC $\rightarrow$ SDH-NA neuron terminals in our slice conditions. The fluorescence intensity increased in a stimulation period-dependent manner (**Figure 2C**). Next, to determine the target cells and receptors of LC $\rightarrow$ SDH-NA neurons, we focused on α_{1A} Rs in astrocytes because our previous study showed that NA increases [Ca²⁺]_i in astrocytes via α_{1A} Rs²². Using spinal cord slices from *NET-Cre* mice that had been injected with AAV-flex[ChrimsonR-tdTomato] into the LC and AAV-gfaABC₁D-GCaMP6m into the SDH (**Figure 2A**), we found that optogenetic stimulation of LC $\rightarrow$ SDH-NA axons/terminals evoked a rise in [Ca²⁺]_i in astrocytes (**Figure 2, D** and **E**). Pretreatment with the α_{1A} R-specific antagonist silodosin (40 nM) suppressed the Ca²⁺ responses (**Figure 2F**), suggesting that NA released from LC $\rightarrow$ SDH-NA neurons induces Ca²⁺ responses in SDH astrocytes via α_{1A} Rs. Behaviorally, intrathecal injection of silodosin in *NET-Cre*;AAV-ChrimsonR mice attenuated mechanical hypersensitivity by optogenetic stimulation of the LC $\rightarrow$ SDH-NA neurons (**Figure 2G**). Consistent with our previous findings²², intrathecal administration of NA (0.1 nmol) evoked transient mechanical hypersensitivity, which was completely abolished in *Hes5-CreERT2;Adra1a^{flox/flox}* mice treated with tamoxifen (Hes5 $^{+}$ astrocyte-selective conditional knockout of α_{1A} Rs: Hes5 $^{+}$ astrocyte- α_{1A} R cKO) (**Figure 2H**). Moreover, mechanical hypersensitivity after acute restraint stress was abolished in Hes5 $^{+}$ astrocyte- α_{1A} R cKO mice (**Figure 2I**). In addition, intrathecal administration of 5,7-dichlorokynurenic acid (DCK; 10 nmol), an antagonist of D-serine signaling on N-methyl-D-aspartate (NMDA) receptors, also suppressed mechanical hypersensitivity (**Figure 2J**). Supporting this finding, our previous report identified D-serine as a hypersensitivity-inducing factor released from Hes5 $^{+}$ astrocytes²². These results indicate that spinally-released NA from LC $\rightarrow$ SDH-NA neurons after acute restraint stress activates α_{1A} Rs on Hes5 $^{+}$ astrocytes, which causes mechanical pain hypersensitivity.

Hes5 $^{+}$ astrocytes mask IN function via adenosine A₁ receptors

Besides astrocytes, *Adra1a* mRNA is also expressed in INs in the SDH^{13,14}. However, *Vgat-Cre;Adra1a^{flox/flox}* mice (*Vgat* $^{+}$ IN-selective conditional knockout of α_{1A} Rs: *Vgat* $^{+}$ IN- α_{1A} R cKO mice)^{14,16} had no effect on the NA (0.1 nmol)-induced mechanical hypersensitivity (**Figure 3A**). A similar behavioral phenotype was also observed after intrathecal administration of phenylephrine (0.05 nmol), a selective agonist for α_1 Rs (**Figure 3B**). Furthermore, stress-induced mechanical hypersensitivity was not affected in *Vgat* $^{+}$ IN- α_{1A} R cKO mice (**Figure 3C**). Considering previous data that SDH-INs (including *Vgat* $^{+}$) respond to NA via α_{1A} Rs at NA concentrations similar to those that elicit Hes5 $^{+}$ astrocyte response^{16,22}, it remains unclear why intrathecally administered NA and spinally released NA from LC $\rightarrow$ SDH-NA neuron terminals (after acute restraint stress) have no suppressive effect via α_{1A} R-mediated *Vgat* $^{+}$ IN activation on mechanical hypersensitivity. We then hypothesized an interaction between Hes5 $^{+}$ astrocytes and *Vgat* $^{+}$ INs as per a previous report that optogenetic activation of rat SDH astrocytes leads to ATP release; ATP is converted to adenosine extracellularly which subsequently suppresses the activity of SDH-INs via adenosine A₁ receptors (A₁Rs)⁴⁵. By performing whole-cell current-clamp recordings of *Vgat* $^{+}$ INs in spinal cord slices from *Vgat-Cre;Rosa-tdTomato* mice, we found that bath application of NA (20 μ M) evoked depolarization in 22.2% (10/45 cells) of tdTomato $^{+}$ cells (*Vgat* $^{+}$

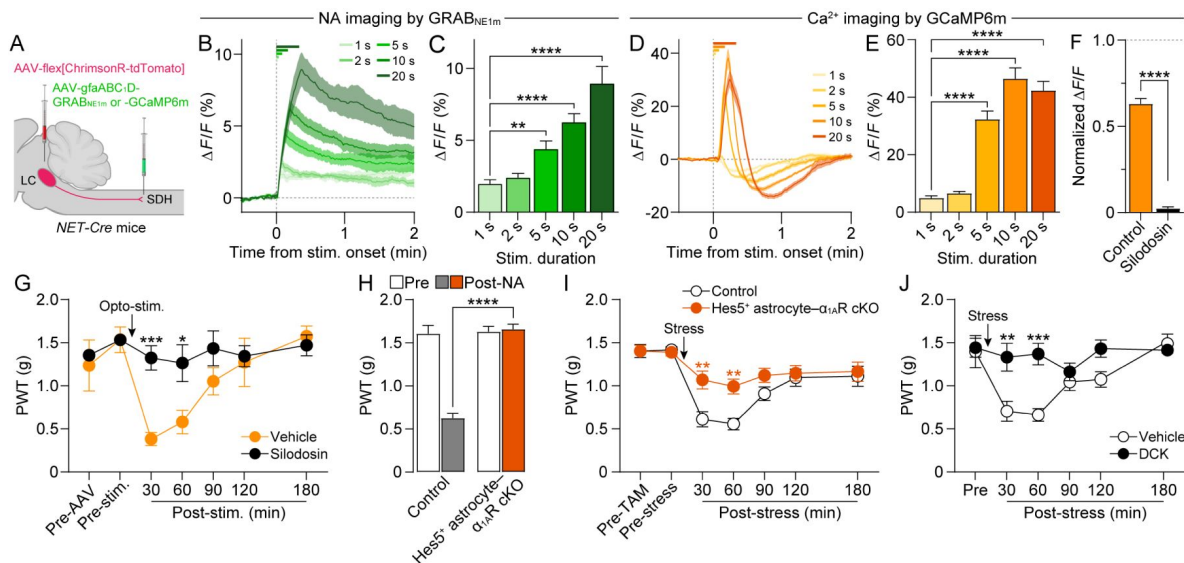


Figure 2.

α_1 ARs in Hes5⁺ SDH astrocytes are required for stress-induced mechanical hypersensitivity

(A) Schematic illustration of intra-SDH microinjection of AAV-gfaABC1D-GRAB_{NE1m} or -GCaMP6m and intra-LC microinjection of AAV-flex[ChrimsonR-tdTomato] in *NET-Cre* mice. (B) Representative traces of GRAB_{NE1m} signals by fluorescence imaging using spinal cord slices. Each trace represents the GRAB_{NE1m} signal before and after optogenetic stimulation (625 nm, 1 mW, 10 Hz, 5 ms pulse duration, 1–20 s). (C) Quantitative analysis of the peak amplitude of GRAB_{NE1m} $\Delta F/F$ after optogenetic stimulation in LC $\rightarrow$ SDH-NA axons/terminals ($n = 4$ slices; one-way ANOVA with post hoc Dunnett's multiple comparisons test; $**P < 0.01$, $****P < 0.0001$). (D) Representative traces of astrocytic GCaMP6m signals by fluorescence imaging using spinal cord slices. Each trace represents the GCaMP6m signal before and after optogenetic stimulation (as described in B). (E) Quantitative analysis of the peak amplitude of GCaMP6m $\Delta F/F$ after optogenetic stimulation in LC $\rightarrow$ SDH-NA axons/terminals ($n = 133$ cells, 4 slices, 4 mice; Friedman test with post hoc Dunn's multiple comparisons test; $****P < 0.0001$). (F) Effect of silodosin (40 nM) on astrocytic Ca^{2+} responses in the SDH after optogenetic stimulation (10 s) in LC $\rightarrow$ SDH-NA axons/terminals (Control, $n = 83$ cells, 4 slices, 4 mice; Silodosin, $n = 53$ cells, 4 slices, 4 mice; Mann-Whitney test; $****P < 0.0001$). (G) Effect of intrathecal silodosin (3 nmol) on mechanical hypersensitivity induced by optogenetic stimulation in LC $\rightarrow$ SDH-NA axons/terminals (Vehicle, $n = 5$ mice; Silodosin, $n = 6$ mice; two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $*P < 0.05$, $***P < 0.001$ vs. vehicle group). (H) Change in PWT at 30 min after intrathecal injection of NA (0.1 nmol) in control (*Adra1a*^{flox/flox}) and Hes5⁺ astrocyte-selective α_1 AR conditional knockout mice [*Hes5-CreERT2*; *Adra1a*^{flox/flox} mice treated with tamoxifen (TAM) (Hes5⁺ astrocyte- α_1 AR cKO mice) ($n = 6$ mice per group; two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $****P < 0.0001$). (I and J) Stress-induced mechanical hypersensitivity in Hes5⁺ astrocyte- α_1 AR cKO mice [I: Control (*Adra1a*^{flox/flox}), $n = 7$ mice; Hes5⁺ astrocyte- α_1 AR cKO, $n = 8$ mice] or wild-type mice with intrathecal DCK (10 nmol) (J: $n = 5$ mice per group) (two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $**P < 0.01$, $***P < 0.001$ vs. control or vehicle group). Data represent mean $\pm$ SEM. This figure was created using BioRender.com.

INs), with 30% of these showing action potentials (3/10 cells). The proportion of $Vgat^+$ INs depolarized by NA is similar to that of *Adora1a* mRNA-expressing INs¹⁴. In 50% (5/10 cells) of depolarized $Vgat^+$ INs, additional application of the A_1 -R-selective agonist N6-cyclopentyladenosine (CPA; 1 μ M) suppressed NA-induced depolarization/action potentials (**Figure 3D** and **E**). Supporting this finding, RNAscope *in situ* hybridization detected *Adora1* mRNA fluorescence in $63.5 \pm 6.9\%$ of *Slc32a1* ($Vgat$)⁺ INs ($n = 497$, from three mice) (**Figure 3F**). Because A_1 Rs are also expressed in the dorsal root ganglion neurons⁴⁶, excitatory neurons and microglia⁴⁷ in the SDH, we employed the CRISPR–Cas9 system using AAV vectors to specifically knockdown A_1 Rs in SDH-INs. AAV vectors designed to express the *Staphylococcus aureus* Cas9 (SaCas9)⁴⁸ in a Cre-dependent manner (AAV-flex[*SaCas9*]) and single guide RNA-expression vectors (AAV-flex[mCherry]-U6-sgAdora1 or - flex[mCherry]-U6-sgYFP [as a control]) were co-microinjected into the SDH of *Vgat-Cre* mice (**Figure 3G**). SaCas9 (detected by HA-tag) and mCherry-labeled cells in the SDH were immunolabeled with paired box 2 (PAX2) (**Figure 3G**), a marker of INs⁴⁹. In these SDH- $Vgat^+$ IN-selective A_1 R conditional knockdown mice (SDH- $Vgat^+$ IN- A_1 R cKD mice), the inhibitory effect of CPA on NA-induced depolarization/action potentials was completely abolished in mCherry⁺ neurons (**Figure 3, H** and **I**). Using a chemogenetic approach with modified human muscarinic G_q -protein-coupled receptors (hM3Dq)⁵⁰, we further examined the effect of G_q -stimulated Hes5⁺ astrocytes on inhibitory synaptic transmission. *Hes5-CreERT2* mice were microinjected with AAV-flex[hM3Dq-HA] into the SDH (*Hes5-CreERT2*;AAV-hM3Dq mice). hM3Dq (detected by HA-tag) was expressed in the SDH, and these cells were double-labeled with glial fibrillary acidic protein (GFAP) and SRY-related high-mobility group box 9 (SOX9), markers of astrocytes (**Figure 3J**). Using spinal cord slices from these mice, we performed whole-cell voltage-clamp recordings of substantia gelatinosa (SG) neurons. Bath application of hM3Dq agonist clozapine N-oxide (CNO; 100 μ M) reduced the frequency of spontaneous inhibitory postsynaptic currents (sIPSCs) in SG neurons (**Figure 3, K** and **L**). This effect was prevented by pretreatment with the A_1 R-specific antagonist, 8-cyclopentyl-1,3-dimethylxanthine (CPT; 1 μ M) (**Figure 3, M** and **N**). These results suggest that a signaling pathway from Hes5⁺ astrocytes suppresses the activity of $Vgat^+$ INs via A_1 Rs in the SDH.

Hes5⁺ astrocyte-mediated inhibitory signals to INs affect mechanical hypersensitivity

We investigated whether A_1 Rs in SDH- $Vgat^+$ INs contributed to the mechanical hypersensitivity elicited by intrathecal NA (0.1 nmol). Acute inhibition of spinal A_1 Rs by intrathecal administration of CPT (3 nmol) suppressed NA-induced mechanical hypersensitivity (**Figure 4A**). Similarly, SDH- $Vgat^+$ IN- A_1 R cKD mice showed a significant reduction in the behavioral hypersensitivity by NA (**Figure 4B**), indicating that A_1 Rs in SDH- $Vgat^+$ INs contribute to mechanical hypersensitivity by spinal NA. Furthermore, SDH- $Vgat^+$ IN- A_1 R cKD mice and intrathecal-CPT-pretreated mice exhibited a significant attenuation of the stress-induced mechanical hypersensitivity (**Figure 4, C** and **D**). These results suggest that Hes5⁺ astrocyte-mediated negative control for SDH- $Vgat^+$ INs via A_1 R-mediated signals is crucial for stress-induced mechanical hypersensitivity. Moreover, we investigated the role of this interaction in stress-induced alteration of somatosensory information processing in the SDH using phosphorylated extracellular signal-regulated kinase (pERK), a neuronal nociceptive input marker in superficial SDH neurons⁵¹. Consistent with the results of a previous report⁵², A β fiber stimulation of the hindpaw alone did not increase the number of pERK⁺ neurons in the superficial SDH (laminae I–III using isolectin B4 [IB4], a marker of lamina II) (**Figure 4, E** and **F**). Similarly, acute restraint stress alone did not change the number of pERK⁺ neurons (**Figure 4, E** and **F**). Interestingly, A β fiber stimulation following acute restraint stress exposure significantly increased the number of pERK⁺ neurons in the superficial SDH (**Figure 4, E** and **F**). Intrathecal injection of CPT prevented the stress-induced increase in pERK⁺ neurons (**Figure 4, E** and **F**). These results suggest that the abnormal

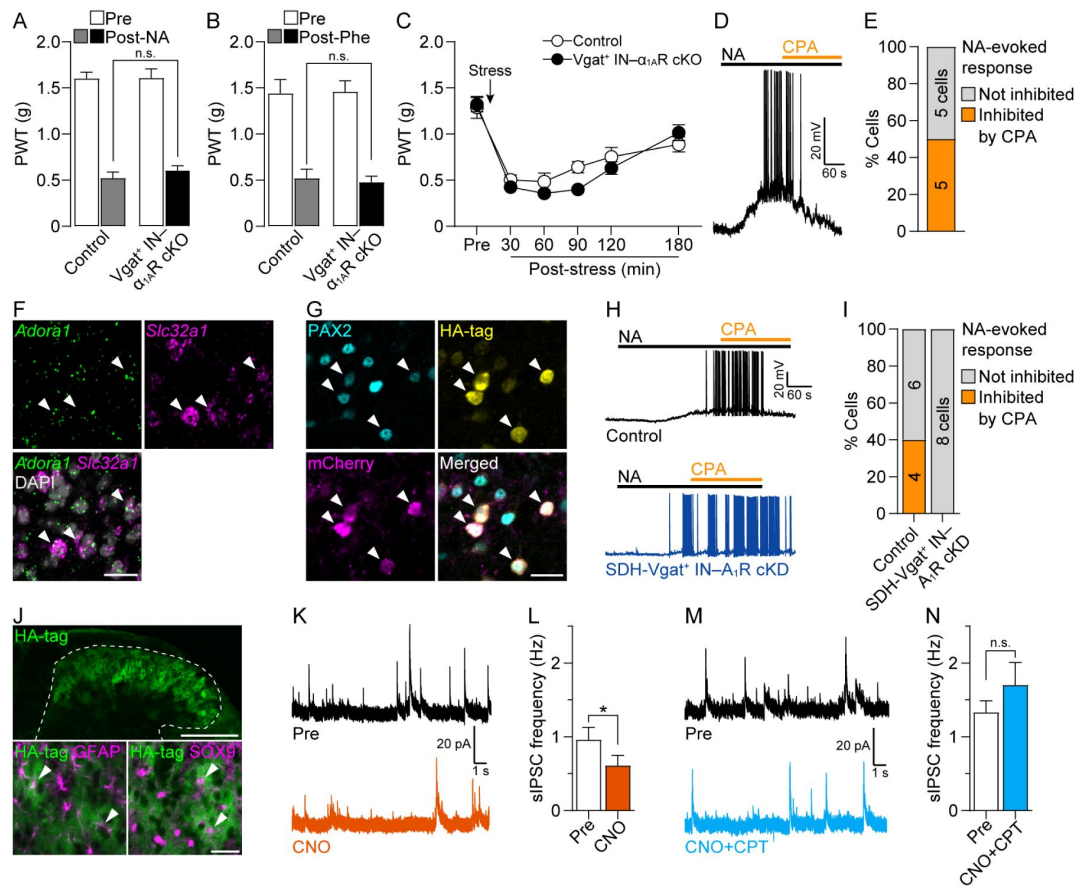


Figure 3.

Activation of *Hes5*⁺ astrocytes reduces activity in *Vgat*⁺ INs in the SDH

(A and B) Intrathecal NA (A, 0.1 nmol) or Phe (B, 0.05 nmol)-induced mechanical hypersensitivity in control (*Adora1*^{flx/flx}) and *Vgat*⁺ IN-selective α_{1A} R conditional knockout mice [*Vgat-Cre;Adora1*^{flx/flx} mice (*Vgat*⁺ INs- α_{1A} R cKO mice)] (*n* = 6 mice per group; two-way ANOVA with post hoc Bonferroni's multiple comparisons test; n.s., not significant). (C) Changes in PWT after acute exposure to restraint stress in control (*Adora1*^{flx/flx}; *n* = 6 mice) and *Vgat*⁺ INs- α_{1A} R cKO mice (*n* = 9 mice) (two-way ANOVA with post hoc Bonferroni's multiple comparisons test). (D) Representative trace of membrane potentials in tdTomato⁺ (*Vgat*⁺) SDH neurons after application of NA (20 μ M) to spinal cord slices from *Vgat-Cre;Rosa-tdTomato* mice. A_1 R agonist CPA (1 μ M) was co-applied with NA. (E) Percentage of *Vgat*⁺ SDH neurons whose NA-evoked response was inhibited by CPA (*n* = 10 cells from 7 mice). (F) Representative images of *Adora1* (A_1 R) mRNA expression (green) in *Slc32a1* (*Vgat*⁺) INs (magenta). DAPI staining is shown in gray. Arrowheads indicate A_1 R-expressing *Vgat*⁺ cells. Scale bar, 25 μ m. (G) SaCas9 (yellow, detected by HA-tag) and mCherry (magenta) expression in the PAX2⁺ INs (cyan) at 3 weeks after intra-SDH injection of AAV-flex[*SaCas9*] and AAV-flex[mCherry]-U6-sg*Adora1* in *Vgat-Cre* mice. Arrowheads indicate genome-editing *Vgat*⁺ cells. Scale bar, 25 μ m. (H) Representative traces of membrane potentials in *Vgat*⁺ INs after application of NA and CPA to spinal cord slices from *Vgat-Cre* mice with conditional knockdown of A_1 Rs in *Vgat*⁺ INs (SDH-*Vgat*⁺ IN- A_1 R cKD) and their controls (Control; *Vgat-Cre* mice with intra-SDH injection of AAV-flex[mCherry]). (I) Percentage of mCherry⁺ (*Vgat*⁺) SDH neurons whose NA-evoked response was inhibited by CPA (Control, *n* = 10 cells from 8 mice; SDH-*Vgat*⁺ IN- A_1 R cKD, *n* = 8 cells from 5 mice). (J) Expression of hM3Dq (green, detected by HA-tag) in the SDH at 3 weeks after intra-SDH injection of AAV-flex[hM3Dq] in *Hes5-CreERT2* mice treated with TAM. Dashed line indicates an outline of the gray matter of SDH. GFAP (magenta, bottom left), SOX9 (magenta, bottom right). Arrowheads indicate HA-tag⁺ astrocytes. Scale bars, 200 μ m (top) and 25 μ m (bottom). (K and L) Representative traces of sIPSCs (K) and quantitative analysis of their frequency (L) in SG neurons in spinal cord slices from *Hes5-CreERT2*;AAV-hM3Dq mice treated with TAM [Pre and CNO: before and after bath application of CNO (100 μ M), respectively] (*n* = 7 cells from 7 mice; Wilcoxon signed-rank test; **P* < 0.05). (M and N) Representative traces of sIPSCs (M) and quantitative analysis of their frequency (N) in SG neurons in spinal cord slices from *Hes5-CreERT2*;AAV-flex[hM3Dq] mice treated with TAM [Pre and CNO: before and after bath application of CNO with CPT (1 μ M), respectively] (*n* = 13 cells from 13 mice; Wilcoxon signed-rank test; n.s., not significant). Data represent mean $\pm$ SEM.

responsiveness of superficial SDH neurons to signals from A β fibers in mice with acute restraint stress, involves adenosine signals that are important for negatively controlling Vgat⁺ INs by NA-stimulated Hes5⁺ astrocytes and for mechanical hypersensitivity.

Discussion

External and internal stressors elicit a variety of biological responses including pain modulation. Acute exposure to a stressor inhibits or facilitates behavioral responses to thermal and mechanical stimuli applied to the skin^{33,38,53}. While the mechanism underlying pain inhibition has been studied³³, the neural basis of pain facilitation by stress remains poorly understood. In this study, we identified a circuit, the descending LC^{→SDH}-NA neurons, that is activated by acute restraint stress exposure and demonstrated that the circuit is necessary for stress-induced pain hypersensitivity. A notable finding of this study was that Hes5⁺ astrocytes, a subpopulation of spinal cord astrocytes²², are the primary target of NA released from descending LC^{→SDH}-NA neurons for mechanical pain hypersensitivity caused by stress. Thus, this top-down signaling pathway from LC^{→SDH}-NA neurons to Hes5⁺ SDH astrocytes is an essential link between stress and pain sensitization.

Spinal NA derived from LC-NA neurons has been classically implicated in pain inhibitory control^{9,11}. The suggested mechanism to this effect is its action on SDH-INs via α_{1A} Rs^{9,16}. In contrast, recent our study has unveiled another facet of spinal NA signaling, which is pronociceptive, by identifying Hes5⁺ astrocytes in the SDH²². Mechanistically, spinal NA acts on α_{1A} Rs expressed on Hes5⁺ astrocytes, increases intracellular Ca²⁺ levels via a G_q/PLC/IP₃ pathway, and causes mechanical hypersensitivity via release of D-serine²², an activator of NMDA receptors. This study showed that the optogenetic activation of LC^{→SDH}-NA neurons produced mechanical hypersensitivity via α_{1A} Rs in SDH-Hes5⁺ astrocytes, indicating the pronociceptive effect of LC^{→SDH}-NAergic signaling. In contrast, it has also been reported that a similar opto-or chemogenetic activation produces an antinociceptive effect on thermal stimuli^{54,55}. The reason for the difference remains unclear; however, the stimulus modality used for the behavioral assay (mechanical vs. thermal) may be responsible, since the SDH circuits for mechanical and thermal information processing have been shown to be distinct^{3,56–59}.

Given that SDH-INs also express α_{1A} Rs^{13,14}, it is unclear why pain hypersensitivity is the preferentially manifested behavioral phenotype after activation of LC^{→SDH}-NAergic signal. This study showed that NA-stimulated Hes5⁺ astrocytes negatively regulate the activity of SDH-Vgat⁺ INs. This astrocytic attenuation of SDH-Vgat⁺ IN activity may involve adenosine signaling because A₁R cKD in SDH-Vgat⁺ INs canceled the suppressive effect of A₁R agonist on NA-induced enhancement of Vgat⁺-IN activity and NA-induced mechanical hypersensitivity. The adenosine-mediated inhibition of SDH-INs is consistent with the data from a previous study⁴⁵. Thus, these results suggest that spinal NA acts on α_{1A} Rs expressed by both Hes5⁺ astrocytes and Vgat⁺ INs; however, NA-stimulated Hes5⁺ astrocytes suppress Vgat⁺ IN activity by adenosine signaling via A₁Rs; this induces a state where the Hes5⁺ astrocyte-derived pain-facilitating effect predominates, causing mechanical pain hypersensitivity (**Figure 4G**).

While a neuronal model for gating peripherally derived sensory signals in the SDH has been previously established by identifying several subpopulations of INs^{1,5}, our study proposes a new regulatory mechanism in this model in which Hes5⁺ astrocytes act as non-neuronal gating cells in the SDH for brain-derived NAergic signaling. This astrocytic gating control of INs may affect processing and transmission of somatosensory information in the SDH. Indeed, a pharmacological blockade of the adenosine signal to SDH-Vgat⁺ INs suppressed the A β fiber-mediated pERK expression in the superficial SDH of stressed mice. Preproenkephalin⁺ IN population may be among the SDH-Vgat⁺ INs involved in stress-induced pain modulation³⁵. Interfering with GABA/glycine signaling in the SDH results in abnormal synaptic inputs from

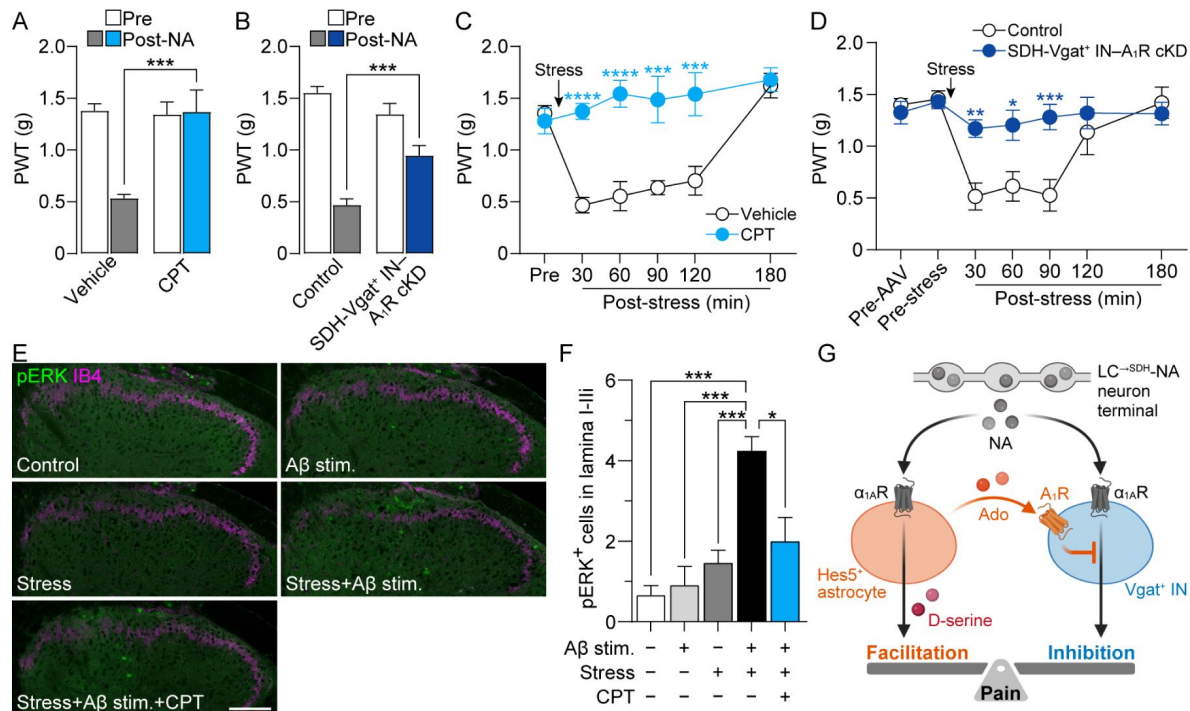


Figure 4.

Hes5⁺ astrocyte-mediated inhibitory signals to SDH-Vgat⁺ INs contribute to stress-induced mechanical hypersensitivity

(A and B) PWT before and 30 min after intrathecal administration of NA (0.1 nmol) in wild-type mice pretreated intrathecally with vehicle or CPT (3 nmol) (A: $n = 5$ mice per group) or in control (*Vgat-Cre* mice with intra-SDH of AAV-flex[mCherry]) and SDH-Vgat⁺ IN-A₁R cKD mice (B: $n = 9$ mice per group) (two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $***P < 0.001$). (C and D) PWT before and after acute restraint stress in wild-type mice pretreated intrathecally with vehicle or CPT (C: Vehicle, $n = 6$ mice; CPT, $n = 5$ mice) or in control (*Vgat-Cre* mice with intra-SDH of AAV-flex[mCherry]) and SDH-Vgat⁺ IN-A₁R cKD mice (D: Control, $n = 6$ mice; SDH-Vgat⁺ IN-A₁R cKD, $n = 7$ mice) (two-way ANOVA with post hoc Bonferroni's multiple comparisons test; $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$ vs. vehicle or control group). (E) Representative images of pERK (green) and IB4 (magenta) immunofluorescence in the SDH with or without Aβ fiber stimulation and/or restraint stress. CPT was intrathecally administered 30 min before stress exposure. (F) Quantitative analysis of the number of pERK⁺ cells in superficial laminae of the SDH in each group ($n = 4-5$ mice per group; one-way ANOVA with post hoc Tukey's multiple comparisons test; $*P < 0.05$, $***P < 0.001$). (G) Schematic illustration of the mechanisms of stress-induced pain facilitation highlighting NA signals from LC-SDH⁺ NAergic terminals to Hes5⁺ astrocytes and Vgat⁺ INs. Data represent mean $\pm$ SEM. This figure was created using BioRender.com.

primary afferent A β fibers onto superficial SDH neurons⁶⁰. Our view is also supported by previous findings from *in vivo* patch-clamp recordings, which show that, in about a half of SG neurons tested, NA enhances excitatory synaptic responses evoked by tactile stimuli (air puff to the skin)⁶¹. Additionally, our hypothesis that Hes5⁺ astrocytes co-release D-serine and ATP/adenosine in response to α_1 A_R activation by NA is supported by previous findings in the neocortex⁶². A β fiber inputs into neurokinin 1 receptor-positive projection neurons in SDH lamina I following inhibition of GABA/glycine signaling is canceled by a pharmacological blockade of NMDA receptors⁶⁰, supporting our data that the stress-induced mechanical hypersensitivity was suppressed by the antagonist of D-serine site of NMDA receptors DCK. However, the effect of astrocytic adenosine on spinal pain processing has been controversial: both, pronociceptive (consistent with our findings)⁴⁵ and antinociceptive⁶³. Such different effects might be related in part to the regionally restricted astrocytic subpopulations and factors for their activation. Pro- and anti-nociceptive effects have been reported to be mediated by astrocytes located in superficial and deeper laminae and stimulated by NA (via G protein-coupled α_1 A_Rs)²² and ATP (via ionotropic P2X7 receptors)⁶³, respectively. In particular, the regional differences in the role of astrocytes has been demonstrated by our previous study in which a selective chemogenetic stimulation of astrocytes in deeper SDH did not cause mechanical hypersensitivity²². Thus, it is conceivable that locally released adenosine from NA-stimulated Hes5⁺ astrocytes, following acute restraint stress, may suppress SDH-Vgat⁺ IN function, which results in mechanical pain hypersensitivity. The spatially restricted neuron–astrocyte communication needs to be investigated in future studies.

In this study, we focused on the mechanisms of mechanical hypersensitivity by acute exposure to restraint stress. Consistent with a recent study³⁸, acute restraint stress caused mechanical hypersensitivity; however, there are also reports showing a reduction in behavioral responses to thermal or mechanical stimuli upon exposure to similar restraint stress^{33,35}. A precise explanation for the bidirectional outcomes remains elusive. However, it is widely recognized that the extent and direction of pain modulation may be due to the nature, duration, and intensity of the stressor^{33,34}. Indeed, while acute and short-term exposure (≤ 1 hour) to restraint stress induces mechanical pain hypersensitivity, longer exposure (2 hours) failed to exhibit the behavioral response (**Figure S1**). It has also been reported that behavioral response to nociceptive thermal stimuli is suppressed by a 2-hour exposure to restraint stress¹⁶. Thus, descending LC–SDH-NA neurons, Hes5⁺ SDH astrocytes, or their output signals may be attenuated by unidentified inhibitory signals evoked by longer exposure to restraint stress. In addition, a similar bidirectional behavioral outcome on mechanical pain has also been observed with intrathecal NA; low doses of NA induce mechanical pain hypersensitivity while higher doses do not²². Thus, the bidirectional change may depend on the level of NA neuronal activity and NA content in the SDH, which is supported by a previous report⁶⁴.

Clinically, stressor-evoked psychological and physical responses are considered critical for the development of chronic pain³⁶, especially nociplastic pain, a condition that arises from altered nociceptive signaling without actual or threatened tissue damage, or disease or lesion of the somatosensory system³⁷. In fact, patients with chronic pain have a high incidence of stress-related psychiatric disorders, such as anxiety, fear, and depression^{11,65}; stressful life events are known to be a key factor in establishing fibromyalgia and enhance pain in patients with irritable bowel syndrome and headaches³⁷. Thus, our findings provide a mechanistic insight into the link between stress and pain modulation and may help in understanding and developing a new strategy for chronic pain, including nociplastic pain.

Methods

Animals

Male C57BL/6J mice (CLEA Japan), male and female *NET-Cre* mice [Tg(*Slc6a2-cre*)#Stl] (kindly provided by Prof. Thomas McHugh, RIKEN Center for Brain Science)³⁹, *Hes5-CreERT2* mice [Tg(*Hes5-cre/ERT2*)^{2Vtr}] (kindly provided by Prof. Verdon Taylor)⁶⁶, *Adra1a*^{flox/flox} mice²², *Vgat-Ires-Cre* mice [*Slc32a1*^{tm2(cre)Lowl}] (Stock No: 016962, The Jackson Laboratory)⁶⁷, and *Rosa26-tdTomato* mice [B6.Cg-Gt(*ROSA*)26Sor^{tm14}(CAG-tdTomato)Hze/J] (Stock No: 007914, The Jackson Laboratory)⁶⁸ were used. To generate *Hes5-CreERT2;Adra1a*^{flox/flox} mice²² or *Vgat-Cre;Adra1a*^{flox/flox} mice¹⁶, *Adra1a*^{flox/flox} mice were crossed with *Hes5-CreERT2* mice or *Vgat-Ires-Cre* (*Vgat-Cre*) mice, and the obtained *Hes5-CreERT2;Adra1a*^{flox/+} mice or *Vgat-Cre;Adra1a*^{flox/+} mice were further crossed with *Adra1a*^{flox/flox} mice. Cre-negative *Adra1a*^{flox/flox} littermate mice were used as controls. For induction of CreERT2 recombinase activity, *Hes5-CreERT2* mice were given an intraperitoneal (i.p.) injection of tamoxifen (#T5648, Sigma-Aldrich; 2 mg dissolved in 100 μ l corn oil [#032-17016; Wako, Saitama, Japan]) once a day for 5–10 successive days. We used tamoxifen-injected mice for further analyses 7 days or more after the last injection. All mice used were 8–12 weeks of age at the start of each experiment and were housed at temperature and humidity ranges of 21–23°C and 40–60%, respectively, with a 12-hour light-dark cycle. All animals were fed food and water ad libitum. All animals were housed in standard polycarbonate cages in groups of same-sex littermates. All animal experiments were conducted according to relevant national and international guidelines contained in the ‘Act on Welfare and Management of Animals’ (Ministry of Environment of Japan) and ‘Regulation of Laboratory Animals’ (Kyushu University) and under the protocols approved by the Institutional Animal Care and Use committee review panels at Kyushu University.

Restraint stress model

According to the methods described in our previous study¹⁶, mice were restrained by placing them in a Falcon® 50-ml conical tube (#352070; Corning) with a hole in the tip. The space remaining behind the mouse was filled with a paper towel. Mice were allowed to breathe but not turn around. Control mice were handled briefly and placed in home cages without water or food for 1 hour.

Measurement of behavioral response to mechanical stimuli using von Frey filaments

Mice were placed individually in an opaque plastic cylinder, which was placed on a wire mesh, and habituated for 0.5–1 hour to allow acclimatization to the new environment. After that, calibrated von Frey filaments (0.02–2.0 g; #NC12775; North Coast Medical) were applied to the plantar surface of the hindpaw of mice from below the mesh floor. 50% paw withdrawal threshold was calculated using the up-down method⁶⁹. The basal mechanical threshold was determined by performing von Frey test before restraint stress exposure, AAV microinjection, DTX administration, optogenetic stimulation, intrathecal injection of several drugs and tamoxifen administration. Mechanical sensitivity was evaluated at 30, 60, 90, 120 and 180 min after restraint stress or optogenetic stimulation, or at 30 min after intrathecal injection of NA or Phe.

Recombinant AAV (rAAV) vector production

pAAV-CAG-flex[GCaMP6s]-WPRES (#100842) and pAAV-synapsin (Syn)-flex[ChrimsonR-tdTomato]-WPRES (#62723) were purchased from Addgene. The genes encoding GCaMP6m (#40754; Addgene), Cre, and GRAB_{NE1m} (#123308; Addgene) were subcloned into the pENTR plasmid (Thermo Fisher Scientific). To produce AAV vectors, we inserted GCaMP6m and GRAB_{NE1m} into pZac2.1-gfaABC₁D-WPRE plasmid and Cre into pZac2.1-enhanced synapsin (ESYN)-WPRES plasmid. To produce the AAV vector for the Cre-switch system, vectors containing the promoters encoding EF1 α were generated from pAAV-CA-FLEX (#38042; Addgene) by substituting the promoter. We then inserted AcGFP, DTR-EGFP (kindly provided by Prof. Kenji Kohno, Nara Institute of Science and

Technology), mCherry and hM3Dq (#45547; Addgene) into pAAV-EF1 α -FLEX. The genes encoding U6-sgRNA (#61593; Addgene) and SaCas9 (#78601; Addgene) were subcloned into the pENTR plasmid. Synthetic oligonucleotides including the targeting sequence for *Adora1* (5'-AAGTTCCGGGTCACCTTTCTG-3') and non-targeting sequence (5'-CCATGTGATCGCGCTTCTCGT-3') were replaced with the targeting site in the original pENTR-U6-sgBsa1 plasmid. The resulting U6-sgRNA cassette was transferred into pAAV-CMV-flex[mCherry]-WPRE plasmid to generate pAAV-CMV-flex[mCherry]-WPRE-U6-sgAdora1/sgYFP (sgControl). The SaCas9 cassette was transferred into pAAV-CMV-flex-WPRE plasmid to generate pAAV-CMV-flex[SaCas9]-WPRE. rAAV vectors were produced from human embryonic kidney 293T (HEK293T) cells with triple transfection [pZac or pAAV, cis plasmid; pAAV2/5 (University of Pennsylvania Gene Therapy Program Vector Core), pAAV2/9 (University of Pennsylvania Gene Therapy Program Vector Core) or pAAV2/retro (#81070; Addgene), trans plasmid; pAd DeltaF6, adenoviral helper plasmid (University of Pennsylvania Gene Therapy Program Vector Core)] and purified by two cesium chloride density gradient purification steps. The vector was dialyzed against phosphate-buffered saline (PBS) containing 0.001% (v/v) Pluronic-F68 (#24040032; Thermo Fisher Scientific) using Vivaspin Turbo 15 100,000 MWCO (#VS15T41; Sartorius). The genome titer of rAAV was determined by Pico Green fluorometric reagent (#P7589; Thermo Fisher Scientific) following denaturation of the AAV particles. Vectors were stored at -80°C until use.

Intra-SDH and intra-LC injection of rAAV vectors

Viral injection was performed in accordance with our previously described method^{22,70}. Mice were deeply anesthetized by subcutaneous injection of ketamine (100 mg/kg) and xylazine (10 mg/kg). For intra-SDH injection, the skin was incised at Th11–L4 vertebrae, and custom-made clamps were attached to the caudal sites of the vertebral column. Paraspinal muscles around the left side of the interspace between Th13 and L1 vertebrae were removed, and the dura mater and arachnoid membrane were carefully incised using the tip of a 30-G needle to make a small window allowing a glass microcapillary to insert directly into the SDH. The microcapillary was inserted into the unilateral SDH (around 120–150 μ m in depth from the surface of the dorsal root entry zone). rAAV solutions (approximately 500 nl) were injected using a Micro4 Micro Syringe Pump Controller (World Precision Instrument). After microinjection, the glass microcapillary was removed, the skin was sutured with 5-0 silk, and the mice were kept on a heating pad until recovery. For intra-LC injection, rAAV solutions were injected (approximately 300 nl in one site) adjacent to the LC (anteroposterior (AP): -5.4 mm from the bregma; mediolateral (ML): $\pm$ 0.9 mm; dorsoventral (DV): -3.2 mm from the dura). Only AAV2/9-CAG-flex[GCaMP6s]-WPRE was injected unilaterally, and the other AAV vectors were injected bilaterally. To keep the bregma and lambda in the same horizontal plane, tolerance was maintained at <50 μ m in the dorsoventral axis between the bregma/lambda. The microcapillary was withdrawn 3 min after ending the injection. The skin was then sutured with 5-0 silk, and the mice were kept on a heating pad until recovery. We used virus-injected mice for further analyses 3 weeks or more after the last injection. The following viral titers were used: AAV2/9-CAG-flex[GCaMP6s]-WPRE, AAV2/9-gfaABC₁D-GCaMP6m-WPRE, AAV2/9-EF1 α -flex[AcGFP]-WPRE (for *NET-Cre* mice), AAV2/9-EF1 α -flex[DTR-EGFP]-WPRE (for *NET-Cre* mice), AAV2/9-EF1 α -flex[mCherry], AAV2/9-hSyn-flex[ChrimsonR-tdTomato], AAV2/9-gfaABC₁D-GRAB_{NE1m}-WPRE, AAV2/5-EF1 α -flex[hM3Dq] and AAV2/9-CMV-flex[SaCas9]-WPRE: 1.0×10^{12} genome copies (GC)/ml; AAV2/retro-ESYN-Cre-WPRE: 3.0×10^{12} GC/ml; AAV2/9-EF1 α -flex[AcGFP] (for WT mice) and AAV2/9-EF1 α -flex[DTR-EGFP]-WPRE (for WT mice): 2.0×10^{12} GC/ml; AAV2/9-CMV-flex[mCherry]-WPRE-U6-sgYFP and AAV2/9-CMV-flex[mCherry]-WPRE-U6-sgAdora1: 0.5×10^{12} GC/ml.

Immunohistochemistry

Mice were deeply anesthetized with an i.p. injection of pentobarbital and transcardially perfused with PBS (#041-20211; Wako, Saitama, Japan) followed by ice-cold 4% paraformaldehyde (PFA; #162-16065; Wako, Saitama, Japan)/PBS. The transverse L4 segments of the spinal cord and brain were removed, postfixed in the same fixative for 3 hours (spinal cord) or overnight (brain) at 4°C,

and placed in 30% sucrose solution for 24–48 hours at 4°C. After incubation, the tissues were embedded in OCT compound (#4583; Sakura Finetek Japan, Osaka, Japan) and stored at –25°C before use. Transverse spinal cord and brain sections (30 µm) were incubated in blocking solution (3% normal goat serum [#S-1000; Vector Laboratories] or normal donkey serum [#017-000-121; Jackson ImmunoResearch]) for 2 hours at room temperature and then incubated for 48 hours at 4°C with primary antibodies: polyclonal rabbit anti-tyrosine hydroxylase (TH; 1:1000; #AB152; Millipore); polyclonal sheep anti-TH (1:1000; #AB1542; Millipore); monoclonal mouse anti-noradrenaline transporter (NET; 1:2000; #NET05-2; Mab Technologies); polyclonal rabbit anti-green fluorescent protein (GFP; 1:1000; #598; MBL International); monoclonal rabbit anti-hemagglutinin (HA)-tag (1:1000; #3724; Cell Signaling); monoclonal rat anti-glia fibrillary acidic protein (GFAP; 1:2000; #13-0300; Invitrogen); polyclonal goat anti-SRY-related high-mobility group box 9 (SOX9; 1:1000; # AF3075; R&D Systems); polyclonal goat anti-paired box 2 (PAX2; 1:500; # AF3364; R&D Systems); monoclonal rat anti-mCherry (1:2000; #M11217; Thermo Fisher Scientific); polyclonal rabbit anti-phospho-p44/42 MAPK (ERK1/2) (Thr202/Tyr204) (pERK; 1:500; #9101; Cell Signaling); anti-isolectin B4 (IB4)-biotin conjugate (1:1000; # I21414; Thermo Fisher Scientific). After incubation, tissue sections were washed and incubated for 3 h at room temperature with secondary antibodies (Alexa Fluor™ 488, 546 and/or 555; #A11001, A11008, A11035, A11056, A11081, S32351; Thermo Fisher Scientific; #ab150178; Abcam; DyLight™ 405; #705-475-147; Jackson ImmunoResearch). Then, tissue sections were washed, slide mounted, and subsequently placed under coverslips with VECTASHIELD Hardmount (Vector Laboratories).

Immunofluorescence images were obtained with confocal laser microscopy (#LSM700; Carl Zeiss).

Fiber photometry

Immediately after microinjection of AAV vectors encoding GCaMPs, the mice were implanted with an optic cannula. Cannulae were purchased from RWD (#R-FOC-BL400C-50NA, Ø1.25 mm ferrule, Ø400 µm optic fiber, 0.50 NA). The length of the optic fiber protruding from the tip of the ferrule was 4.0 mm. Optic cannula was implanted in the unilateral LC (AP: –5.4 mm from the bregma; ML: 0.9 mm; DV: –3.0 mm from the dura). Next, the cannula was secured using two types of dental glues (Super-bond; Sun Medical, Shiga, Japan; UNIFAST III; GC, Tokyo, Japan). After implantation, the skin was sutured with 5-0 silk, and the mice were kept on a heating pad until recovery. We used cannula-implanted mice for fiber photometry 3 weeks or more after the surgery. GCaMP fluorescent signals were obtained with the fiber photometry apparatus from Doric Lenses, which includes a fiber photometry console (FPC_V6), a 415-nm LED illumination (CLED_415), a 465-nm LED illumination (CLED_465), an LED driver (LEDD_2), and a fluorescence mini cube with built-in fluorescence photodetector amplifier (iFMC4-G2_IE(410-420)_E(460-490)_F(500-550)_S). A 415-nm light was modulated at 208 Hz, while a 465-nm light was modulated at 572 Hz. The optic cannula implanted in mice and fluorescence mini cube were connected by a low-autofluorescence patch cable (#MAF1L1; Thorlabs). The power output at the fiber tip was 30–40 µW. After connecting to the patch cable, the mice were allowed to move freely in the home cage for at least 10 min for habituation. After habituation, fluorescence recording was started. Emitted signals from the tissue were collected through the same fiber and sampled at 12 kHz.

Processing and analysis of fiber photometry data

To avoid the effects of photobleaching, data from the first 5 min of recording were discarded. The raw data were then decimated to 30 Hz (down-sampled by 400) and low-pass Butterworth-filtered at 2 Hz. The 415-nm excitation channel served as an isosbestic, Ca²⁺-independent control wavelength for GCaMP, allowing for bleaching and movement artifact corrections when directly fitted to the Ca²⁺-dependent 465-nm channel. The Correction Baseline function from Doric Neuroscience Studio (Doric Lenses) was used to fit the 415-nm signal to the 465-nm signal using an adaptive iterative re-weighted Penalized Least Squares algorithm⁷¹. The fluorescence change was expressed as a relative change, $\Delta F/F = (F_{465} - F_{415})/F_{415}$, where F_{465} is the 465 nm-induced Ca²⁺-dependent signal, and F_{415} is the 415 nm-induced Ca²⁺-independent signal. Robust Z-scores were

then calculated as $Z = (\Delta F/F - \text{Median } \Delta F/F) / \text{NIQR}$, where Median $\Delta F/F$ is the median of $\Delta F/F$ during the entire measurement period, and NIQR is the normalized interquartile range ($0.7413 \times$ interquartile range). Ca^{2+} events were detected with a threshold of Z-score >3 . The frequency of Ca^{2+} events was calculated as the number of Ca^{2+} events per minute in the 10 min before stress, 60 min during stress (10 min $\times$ 6 times), and 10 min after stress.

Drug administration

To ablate DTR⁺ neurons, DTX (10 $\mu\text{g/kg}$ in PBS; #048-34371; Wako, Saitama, Japan) was administered i.p. for two successive days 3 weeks after intra-LC injection of AAV2/9-EF1 α -flex[DTR-EGFP] or [AcGFP]-WPPE. We used DTX-injected mice for further analyses 2 weeks or more after the last injection. For intrathecal injection, a 30-G needle attached to a Hamilton microsyringe was inserted between the L5/L6 vertebrae and then punctured through the dura. The following drugs and doses were used for intrathecal injection experiments: silodosin (3 nmol in 5 μl PBS; #191-17591; Wako, Saitama, Japan; injection was given 30 min before optogenetic stimulation); L-norepinephrine hydrochloride (0.1 nmol in 5 μl saline; #74480; Sigma-Aldrich); 5,7-dichlorokynurenic acid (DCK; 10 nmol in 5 μl saline of 2% dimethylsulfoxide (DMSO); #0286; Tocris; injection was given immediately before the start of exposure to restraint stress); (R)-(-)-phenylephrine hydrochloride (Phe; 0.05 nmol in 5 μl saline; #163-11791; Wako, Saitama, Japan); 8-cyclopentyl-1,3-dimethylxanthine (CPT; 3 nmol in 5 μl saline of 2% DMSO; #C102; Sigma-Aldrich; injection was given immediately before the start of exposure to restraint stress or 30 min before intrathecal injection of NA).

In vivo optogenetic stimulation

Immediately after microinjection of AAV vectors encoding ChrimsonR, mice were implanted with an optic cannula. A cannula comprised a ceramic ferrule (#CFLC230-10; Thorlabs; ϕ 1.25 mm, 6.4 mm length) and an optic fiber (#FT200UMT; Thorlabs; ϕ 200 μm , 0.39 NA). The length of the optic fiber protruding from the tip of the ferrule was less than 0.3 mm. The vertebrae were immobilized in place using a metal bar along the Th12–L1. A small laminectomy was performed through the Th13 bone using a dental drill. The cannula was secured above the dura mater using two types of dental glues (Super-bond; Sun Medical; UNIFAST III; GC). In this procedure, the dura mater and spinal cord parenchyma were left intact and unscathed. After implantation, the skin was sutured with 5-0 silk, and the mice were kept on a heating pad until recovery. We used cannula-implanted mice for optogenetic stimulation 3 weeks or more after the surgery. To activate ChrimsonR in LC- SDH^+ -NA axons/terminals, 625-nm LED light (#M625F2; Thorlabs) was delivered through a ferrule patch cable (#M83L01; Thorlabs). Light stimulations (2 mW, 10 Hz, 5-ms pulse duration, 5-s light on, 15-s light off, 10 cycles) were generated by an LED driver (#LEDD_2; Doric Lenses).

Fluorescent Ca^{2+} and NA imaging in spinal cord slices

Fluorescent imaging was performed in accordance with our previously described method²². GCaMP6m- and GRAB_{NE1m}-expressing mice were anesthetized with an i.p. injection of urethane (1.2–1.5 mg/kg), and lumbosacral laminectomy was performed. The spinal cord (L1–S2) was removed and placed in a cold, high-sucrose, artificial cerebrospinal (aCSF) fluid (27 mM NaHCO_3 , 1.4 mM NaH_2PO_4 , 2.5 mM KCl, 7.0 mM MgSO_4 , 1.0 mM CaCl_2 , 222 mM sucrose, and 0.5 mM ascorbic acid), which was bubbled with 95% O_2 and 5% CO_2 . After cutting all ventral and dorsal roots, transverse L4 spinal cord slices (350 μm thick) were made using a vibrating microtome (#VT1200; Leica); the slices were kept in oxygenated aCSF solution (125 mM NaCl, 1.25 mM NaH_2PO_4 , 2.5 mM KCl, 1.0 mM MgCl_2 , 2.0 mM CaCl_2 , 26 mM NaHCO_3 , and 20 mM glucose) at room temperature (22–25°C) for at least 30 min before use. The slices were placed in a recording chamber, and fluorescent signals were measured using a confocal laser scanning microscope (#FV3000; Olympus, Tokyo, Japan). The chamber was perfused with aCSF saturated with 95% O_2 and 5% CO_2 at 31–32°C and at 2–3 ml/min using peristaltic pumps. A 488-nm diode laser was used for the excitation of GCaMP6m and GRAB_{NE1m}. The size of the image was $509 \times 509 \mu\text{m}^2$ (512×512 pixels).

The recordings were acquired at a rate of 1.083 s per image. Light stimulations (1 mW, 10 Hz, 5-ms pulse duration, 1–20 s) were generated by an LED driver (#LEDD_2; Doric Lenses) and delivered through an optic fiber (Thorlabs, FT200UMT, $\varnothing$ 200 μ m, 0.39 NA). For an experiment on α_1 AR inhibition, optogenetic stimulations (10 s) were performed twice (at least 15-min apart) in the same slice. Silodosin (40 nM in aCSF) was continuously applied by bath application starting 5 min before the onset of 2nd optogenetic stimulation. After optogenetic stimulation, NA (1 or 10 μ M in aCSF) was applied by bath application as the positive control.

Processing and analysis of Ca^{2+} and NA imaging data

Videos were imported into ImageJ Fiji (<https://imagej.net/software/fiji/>), and motion artifacts were corrected using TurboReg⁷². The fluorescence change was expressed as a relative percentage change, $\Delta F/F = 100 \times (F_t - F_0)/F_0$, where F_t is the fluorescence at time t , and F_0 is the baseline average for 30 frames before the start of stimulation. For Ca^{2+} imaging using GCaMP, the size of each region of interest (ROI) was set to a circle with a diameter of 10 μm^{22} . ROIs were manually identified based on the following criteria: located within 200 μ m from the surface of gray matter and $\geq 30\%$ $\Delta F/F$ fluorescence change after optogenetic stimulation or NA application. For the experiment on α_1 AR inhibition, normalized $\Delta F/F$ was calculated as $(\Delta F/F \text{ in 2nd stimulation})/(\Delta F/F \text{ in 1st stimulation})$. For NA imaging using GRAB_{NE1m}, the raw image was binarized, and a single ROI was detected from such a binarized image with the following parameters: contiguous pixel number >100 and $0 < \text{circularity} < 1.64$. Calculation of $\Delta F/F$ in ROIs was performed using the Time Series Analyzer V3 plugin in ImageJ.

Whole-cell patch-clamp recordings using spinal cord slices

According to our previously described method¹⁶, mice were deeply anesthetized with urethane (1.2–1.5 mg/kg, i.p.), and the lumbar spinal cord was removed and placed in a cold high-sucrose aCSF (250 mM sucrose, 2.5 mM KCl, 2 mM CaCl_2 , 2 mM MgCl_2 , 1.2 mM NaH_2PO_4 , 25 mM NaHCO_3 , and 11 mM glucose). Parasagittal and transverse spinal cord slices (250–300 μ m thick) were made with a vibrating microtome (VT1200, Leica) and then the slices kept in oxygenated aCSF solution (125 mM NaCl, 2.5 mM KCl, 2 mM CaCl_2 , 1 mM MgCl_2 , 1.25 mM NaH_2PO_4 , 26 mM NaHCO_3 , and 20 mM glucose) at room temperature (22–25°C) for at least 30 min. The spinal cord slice was then put into a recording chamber, where it was continuously superfused with aCSF solution at 25–28°C at a flow rate of 4–6 ml/min. Recordings were made with the Axopatch 700B amplifier and pCLAMP 10.4 acquisition software (Molecular Devices). Data were digitized with an analog-to-digital converter (Digidata 1550; Molecular Devices), stored on a personal computer with a data acquisition program (ClampeX version 10.4; Molecular Devices), and analyzed with a software package (Clampfit version 10.7; Molecular Devices). Spontaneous inhibitory postsynaptic currents (sIPSCs) were recorded in the voltage-clamp mode at a holding potential of 0 mV. Patch pipettes were filled with an internal solution (120 mM CsMeSO_4 , 15 mM CsCl, 10 mM HEPES, 5 mM QX-314, 4 mM MgATP, 0.3 mM Na_2GTP , 0.2 mM EGTA, 10 mM TEA-Cl and 8 mM NaCl [pH 7.28] adjusted with CsOH), and whole-cell patch-clamp recordings were made from SG neurons. Resting membrane potentials (RMPs) were recorded in current-clamp mode. Patch pipettes were filled with an internal solution (125 mM K-gluconate, 10 mM KCl, 0.5 mM EGTA, 10 mM HEPES, 4 mM ATP-Mg, 0.3 mM NaGTP, 10 mM phosphocreatine, pH 7.28 adjusted with KOH), and whole-cell patch-clamp recordings were made from tdTomato⁺ or mCherry⁺ neurons. The following drugs were used: NA (20 μ M; #74480; Sigma-Aldrich), clozapine-N-oxide (CNO; 100 μ M; #BML-NS105; Enzo Life Sciences), CPT (1 μ M), N6-Cyclopentyladenosine (CPA; 1 μ M; #119135; Sigma-Aldrich), CNQX disodium salt hydrate (CNQX; 10 μ M; #C239; Sigma-Aldrich), (+)-MK-801 hydrogen maleate (MK-801; 20 μ M; #M107; Sigma-Aldrich), 1(S), 9(R)-(-)-bicuculline methiodide (10 μ M; #14340; Sigma-Aldrich) and strychnine (1 μ M; #S0532; Sigma-Aldrich). All drugs were dissolved in aCSF solution. CNO was superfused for 3 min. Bath application of CPT was continuously superfused from 4 min before CNO application. NA was superfused with an antagonist cocktail (CNQX, MK-801, bicuculline and strychnine) for 3–5 min. If the recording neuron was depolarized by NA perfusion, then CPA was co-perfused with NA and an antagonist cocktail for 3 min. The frequency of sIPSCs

for 1 min of pre- and post-NA application were quantified using Clampfit version 10.7 (Molecular Devices). We quantified averaged RMP for 1 min of pre- and post-drug application, and a change in RMP (Δ RMP) of 5 mV or more was judged to be depolarization or hyperpolarization.

***In situ* hybridization**

Mice were deeply anesthetized with an i.p. injection of pentobarbital and transcardially perfused with PBS followed by ice-cold 4% PFA/PBS. The L4 spinal cord was removed and postfixed in the same fixative 24 hours at 4°C. After that, tissues were incubated with 10%, 20% and 30% sucrose solutions at 4°C, embedded in OCT compound, and stored at -25°C before use. The L4 segments were sectioned at a thickness of 14 μ m. *In situ* hybridization was performed using RNAscope® Multiplex Fluorescent Reagent Kit v2 (#323100; ACDbio), according to the manufacture's protocol for fixed frozen tissue. The following probes were used: Mm-Adora1 (#402261; ACDbio); Mm-Slc32a1-C3 (#319191-C3; ACDbio). Tissue sections were analyzed using an LSM700 Imaging System. Cells were considered positive if three or more punctate dots were present in the nucleus and/or cytoplasm.¹⁴

Electrical stimulation of A β fibers and counting of pERK⁺ neurons in the SDH

After exposure to 1-hour restraint stress, mice were deeply anesthetized with an i.p. injection of urethane (1.2–1.5 mg/kg). Thirty min after stress, A β fiber stimulation was performed. The electrodes were attached to the plantar and dorsal surfaces of the left hindpaw. Transcutaneous nerve stimuli were applied using a stimulus generator (#STG4002-16 mA; Multi Channel Systems). The current intensity of the 2,000-Hz stimuli was 1,000 μ A.⁵² Mice were fixed at 2 min after electrical stimulation. For quantification of pERK⁺ cells, three sections from the L4 spinal cord segments were randomly selected from each mouse, and the number of pERK⁺ neurons in the superficial laminae I–III (defined by staining of IB4) was counted. Consistent with a previous study⁵², we confirmed that electrical stimulation of A δ and C fibers, but not A β fibers, induced ERK phosphorylation in SDH neurons (data not shown).

Statistical analysis

Statistical analyses were performed using Prism (GraphPad). Quantitative data were expressed as the mean $\pm$ SEM. Statistical significance of differences was determined by using two-tailed paired *t*-test (Figure. S1), Mann-Whitney test (Figure. 2F), Wilcoxon signed-rank test (Figure. 3L and 3N), Friedman test with Dunn's multiple comparisons test (Figure. 1E and 2E), one-way ANOVA with Dunnett's multiple comparisons test (Figure. 2C), one-way ANOVA with Tukey's multiple comparisons test (Figure. 4F) and repeated measures two-way ANOVA with Bonferroni's multiple comparisons test (Figure. 1B, 1H, 1K, 1N, 2G, 2H, 2I, 2J, 3A, 3B, 3C, 4A, 4B, 4C and 4D). P values are indicated as **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001.

Supplemental Information for

Supplemental Figures

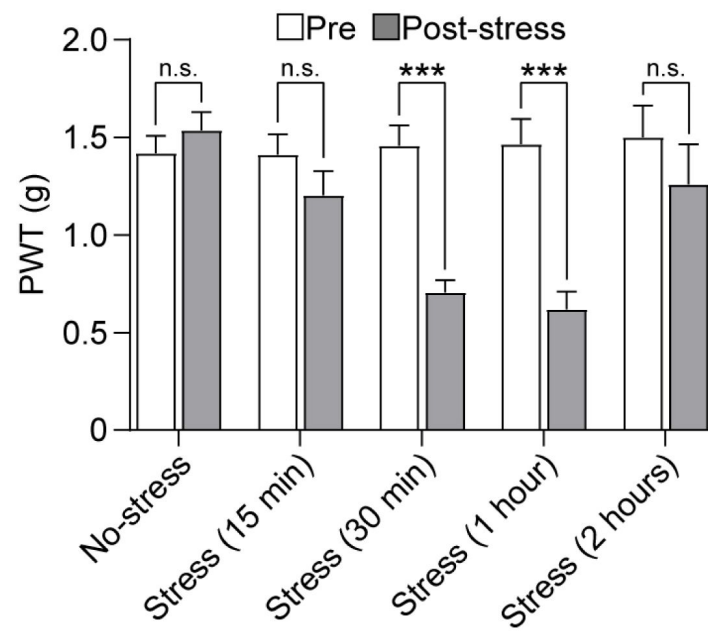


Figure S1.

Mechanical hypersensitivity is weak in mice with longer exposure (2 hours) to restraint stress

PWT change at 30 min after various exposure periods of restraint stress (15 min, 30 min, 1 hour, and 2 hours) in wild-type mice ($n = 5$ mice; two-tailed paired t -test; *** $P < 0.001$; n.s., not significant vs. pre group). Data represent mean $\pm$ SEM.

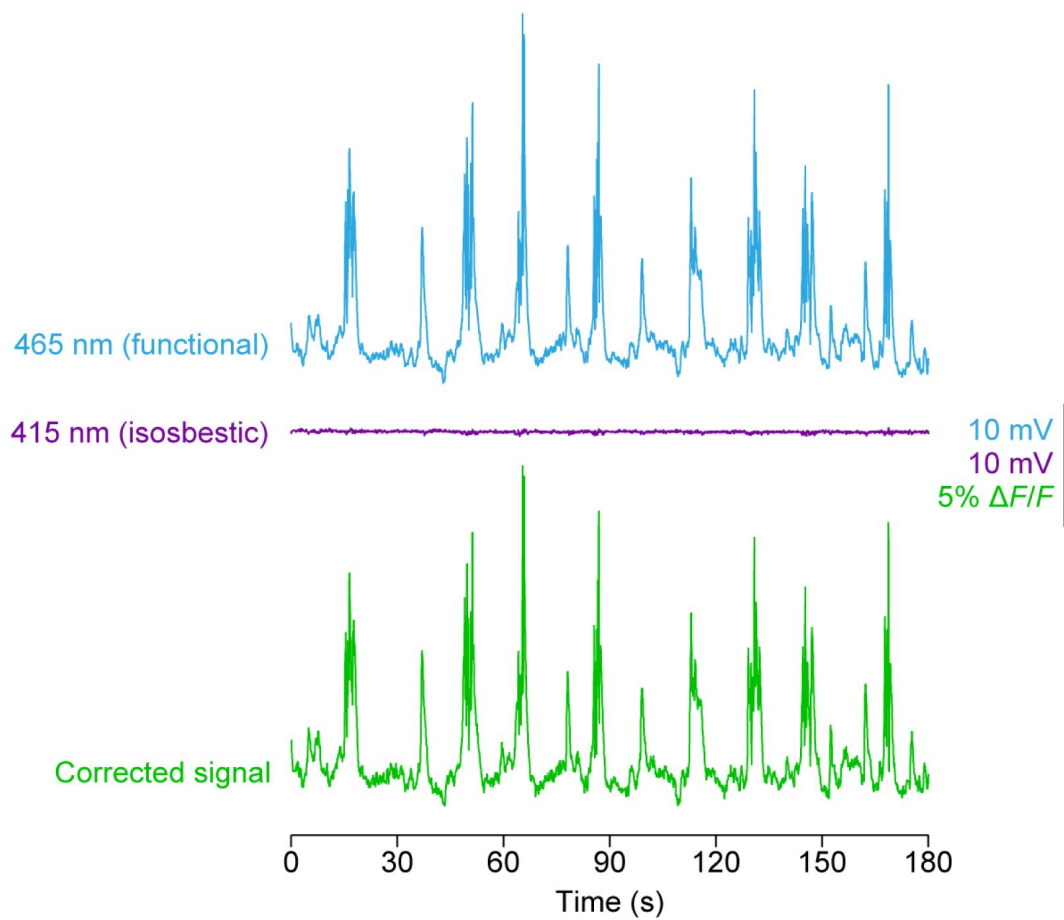


Figure S2.

Raw fluorescent signals in LC-NA neurons during restraint stress (in vivo fiber photometry)

Representative traces of GCaMP6s signals in LC-NA neurons during restraint stress. Traces shown at the top (blue), middle (purple), and bottom (green) indicate 465-nm, 415-nm, and corrected fluorescent signals, respectively.

Data and materials availability

All data associated with this study are present in the paper or the Supplemental Information.

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Reviewer #1 (Public review):**Summary**

In this article, Kawanabe-Kobayashi et al., aim to examine the mechanisms by which stress can modulate pain in mice. They focus on the contribution of noradrenergic neurons (NA) of the locus coeruleus (LC). The authors use acute restraint stress as a stress paradigm and found that following one hour of restraint stress mice display mechanical hypersensitivity. They show that restraint stress causes the activation of LC NA neurons and the release of NA in the spinal cord dorsal horn (SDH). They then examine the spinal mechanisms by which LC → SDH NA produces mechanical hypersensitivity. The authors provide evidence that NA can act on alphaA1Rs expressed by a class of astrocytes defined by the expression of Hes (Hes+). Furthermore, they found that NA, presumably through astrocytic release of ATP following NA action on alphaA1Rs Hes+ astrocytes, can cause an adenosine-mediated inhibition of SDH inhibitory interneurons. They propose that this disinhibition mechanism

could explain how restraint stress can cause the mechanical hypersensitivity they measured in their behavioral experiments.

Strengths:

- (1) Significance. Stress profoundly influences pain perception; resolving the mechanisms by which stress alters nociception in rodents may explain the well-known phenomenon of stress-induced analgesia and/or facilitate the development of therapies to mitigate the negative consequences of chronic stress on chronic pain.
- (2) Novelty. The authors' findings reveal a crucial contribution of Hes⁺ spinal astrocytes in the modulation of pain thresholds during stress.
- (3) Techniques. This study combines multiple approaches to dissect circuit, cellular, and molecular mechanisms including optical recordings of neural and astrocytic Ca²⁺ activity in behaving mice, intersectional genetic strategies, cell ablation, optogenetics, chemogenetics, CRISPR-based gene knockdown, slice electrophysiology, and behavior.

Weaknesses:

- (1) Mouse model of stress. Although chronic stress can increase sensitivity to somatosensory stimuli and contribute to hyperalgesia and anhedonia, particularly in the context of chronic pain states, acute stress is well known to produce analgesia in humans and rodents. The experimental design used by the authors consists of a single one-hour session of restraint stress followed by 30 min to one hour of habituation and measurement of cutaneous mechanical sensitivity with von Frey filaments. This acute stress behavioral paradigm corresponds to the conditions in which the clinical phenomenon of stress-induced analgesia is observed in humans, as well as in animal models. Surprisingly, however, the authors measured that this acute stressor produced hypersensitivity rather than antinociception. This discrepancy is significant and requires further investigation.
- (2) Specifically, is the hypersensitivity to mechanical stimulation also observed in response to heat or cold on a hotplate or coldplate?
- (3) Using other stress models, such as a forced swim, do the authors also observe acute stress-induced hypersensitivity instead of stress-induced antinociception?
- (4) Measurement of stress hormones in blood would provide an objective measure of the stress of the animals.
- (5) Results:
 - a) Optical recordings of Ca²⁺ activity in behaving rodents are particularly useful to investigate the relationship between Ca²⁺ dynamics and the behaviors displayed by rodents.
 - b) The authors report an increase in Ca²⁺ events in LC NA neurons during restraint stress: Did mice display specific behaviors at the time these Ca²⁺ events were observed such as movements to escape or orofacial behaviors including head movements or whisking?
 - c) Additionally, are similar increases in Ca²⁺ events in LC NA neurons observed during other stressful behavioral paradigms versus non-stressful paradigms?
 - d) Neuronal ablation to reveal the function of a cell population.
 - e) The proportion of LC NA neurons and LC → SDH NA neurons expressing DTR-GFP and ablated should be quantified (Figures 1G and J) to validate the methods and permit interpretation of the behavioral data (Figures 1H and K). Importantly, the nocifensive responses and behavior of these mice in other pain assays in the absence of stress (e.g., hotplate) and a few standard assays (open field, rotarod, elevated plus maze) would help

determine the consequences of cell ablation on processing of nociceptive information and general behavior.

f) Confirmation of LC NA neuron function with other methods that alter neuronal excitability or neurotransmission instead of destroying the circuit investigated, such as chemogenetics or optogenetics, would greatly strengthen the findings. Optogenetics is used in Figure 1M, N but excitation of LC → SDH NA neuron terminals is tested instead of inhibition (to mimic ablation), and in naïve mice instead of stressed mice.

g) Alpha1Ars. The authors noted that "Adra1a mRNA is also expressed in INs in the SDH".

h) The authors should comprehensively indicate what other cell types present in the spinal cord and neurons projecting to the spinal cord express alpha1Ars and what is the relative expression level of alpha1Ars in these different cell types.

i) The conditional KO of alpha1Ars specifically in Hes5+ astrocytes and not in other cell types expressing alpha1Ars should be quantified and validated (Figure 2H).

j) Depolarization of SDH inhibitory interneurons by NA (Figure 3). The authors' bath applied NA, which presumably activates all NA receptors present in the preparation.

k) The authors' model (Figure 4H) implies that NA released by LC → SDH NA neurons leads to the inhibition of SDH inhibitory interneurons by NA. In other experiments (Figure 1L, Figure 2A), the authors used optogenetics to promote the release of endogenous NA in SDH by LC → SDH NA neurons. This approach would investigate the function of NA endogenously released by LC NA neurons at presynaptic terminals in the SDH and at physiological concentrations and would test the model more convincingly compared to the bath application of NA.

l) As for other experiments, the proportion of Hes+ astrocytes that express hM3Dq, and the absence of expression in other cells, should be quantified and validated to interpret behavioral data.

m) Showing that the effect of CNO is dose-dependent would strengthen the authors' findings.

n) The proportion of SG neurons for which CNO bath application resulted in a reduction in recorded sIPSCs is not clear.

o) A1Rs. The specific expression of Cas9 and guide RNAs, and the specific KD of A1Rs, in inhibitory interneurons but not in other cell types expressing A1Rs should be quantified and validated.

(6) Methods:

It is unclear how fiber photometry is performed using "optic cannula" during restraint stress while mice are in a 50ml falcon tube (as shown in Figure 1A).

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Reviewer #2 (Public review):

Summary:

This study investigates the role of spinal astrocytes in mediating stress-induced pain hypersensitivity, focusing on the LC (locus coeruleus)-to-SDH (spinal dorsal horn) circuit and its mechanisms. The authors aimed to delineate how LC activity contributes to spinal astrocytic activation under stress conditions, explore the role of noradrenaline (NA) signaling

in this process, and identify the downstream astrocytic mechanisms that influence pain hypersensitivity.

The authors provide strong evidence that 1-hour restraint stress-induced pain hypersensitivity involves the LC-to-SDH circuit, where NA triggers astrocytic calcium activity via alpha1a adrenoceptors (alpha1aRs). Blockade of alpha1aRs on astrocytes - but not on Vgat-positive SDH neurons - reduced stress-induced pain hypersensitivity. These findings are rigorously supported by well-established behavioral models and advanced genetic techniques, uncovering the critical role of spinal astrocytes in modulating stress-induced pain.

However, the study's third aim - to establish a pathway from astrocyte alpha1aRs to adenosine-mediated inhibition of SDH-Vgat neurons - is less compelling. While pharmacological and behavioral evidence is intriguing, the ex vivo findings are indirect and lack a clear connection to the stress-induced pain model. Despite these limitations, the study advances our understanding of astrocyte-neuron interactions in stress-pain contexts and provides a strong foundation for future research into glial mechanisms in pain hypersensitivity.

Strengths:

The study is built on a robust experimental design using a validated 1-hour restraint stress model, providing a reliable framework to investigate stress-induced pain hypersensitivity. The authors utilized advanced genetic tools, including retrograde AAVs, optogenetics, chemogenetics, and subpopulation-specific knockouts, allowing precise manipulation and interrogation of the LC-SDH circuit and astrocytic roles in pain modulation. Clear evidence demonstrates that NA triggers astrocytic calcium activity via alpha1aRs, and blocking these receptors effectively reduces stress-induced pain hypersensitivity.

Weaknesses:

Despite its strengths, the study presents indirect evidence for the proposed NA-to-astrocyte(alpha1aRs)-to-adenosine-to-SDH-Vgat neurons pathway, as the link between astrocytic adenosine release and stress-induced pain remains unclear. The ex vivo experiments, including NA-induced depolarization of Vgat neurons and chemogenetic stimulation of astrocytes, are challenging to interpret in the stress context, with the high CNO concentration raising concerns about specificity. Additionally, the role of astrocyte-derived D-serine is tangential and lacks clarity regarding its effects on SDH Vgat neurons. The astrocyte calcium signal "dip" after LC optostimulation-induced elevation are presented without any interpretation.

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Reviewer #3 (Public review):

Summary

This is an exciting and timely study addressing the role of descending noradrenergic systems in nocifensive responses. While it is well-established that spinally released noradrenaline (aka norepinephrine) generally acts as an inhibitory factor in spinal sensory processing, this system is highly complex. Descending projections from the A6 (locus coeruleus, LC) and the A5 regions typically modulate spinal sensory processing and reduce pain behaviours, but certain subpopulations of LC neurons have been shown to mediate pronociceptive effects, such as those projecting to the prefrontal cortex (Hirshberg et al., PMID: 29027903).

The study proposes that descending cerulean noradrenergic neurons potentiate touch sensation via alpha-1 adrenoceptors on Hes5+ spinal astrocytes, contributing to mechanical hyperalgesia. This finding is consistent with prior work from the same group (dd et al., PMID:). However, caution is needed when generalising about LC projections, as the locus coeruleus is functionally diverse, with differences in targets, neurotransmitter co-release, and behavioural effects. Specifying the subpopulations of LC neurons involved would significantly enhance the impact and interpretability of the findings.

Strengths

The study employs state-of-the-art molecular, genetic, and neurophysiological methods, including precise CRISPR and optogenetic targeting, to investigate the role of Hes5+ astrocytes. This approach is elegant and highlights the often-overlooked contribution of astrocytes in spinal sensory gating. The data convincingly support the role of Hes5+ astrocytes as regulators of touch sensation, coordinated by brain-derived noradrenaline in the spinal dorsal horn, opening new avenues for research into pain and touch modulation.

Furthermore, the data support a model in which superficial dorsal horn (SDH) Hes5+ astrocytes act as non-neuronal gating cells for brain-derived noradrenergic (NA) signalling through their interaction with substantia gelatinosa inhibitory interneurons. Locally released adenosine from NA-stimulated Hes5+ astrocytes, following acute restraint stress, may suppress the function of SDH-Vgat+ inhibitory interneurons, resulting in mechanical pain hypersensitivity. However, the spatially restricted neuron-astrocyte communication underlying this mechanism requires further investigation in future studies.

Weaknesses

(1) Specificity of the LC Pathway targeting

The main concern lies with how definitively the LC pathway was targeted. Were other descending noradrenergic nuclei, such as A5 or A7, also labelled in the experiments? The authors must convincingly demonstrate that the observed effects are mediated exclusively by LC noradrenergic terminals to substantiate their claims (i.e. "we identified a circuit, the descending LC → SDH-NA neurons").

a) For instance, the direct vector injection into the LC likely results in unspecific effects due to the extreme heterogeneity of this nucleus and retrograde labelling of the A5 and A7 nuclei from the LC (i.e., Li et al., PMID: 26903420).

b) It is difficult to believe that the intersectional approach described in the study successfully targeted LC → SDH-NA neurons using AAVrg vectors. Previous studies (e.g., PMID: 34344259 or PMID: 36625030) demonstrated that similar strategies were ineffective for spinal-LC projections. The authors should provide detailed quantification of the efficiency of retrograde labelling and specificity of transgene expression in LC neurons projecting to the SDH.

c) Furthermore, it is striking that the authors observed a comparably strong phenotypical change in Figure 1K despite fewer neurons being labelled, compared to Figure 1H and 1N with substantially more neurons being targeted. Interestingly, the effect in Figure 1K appears more pronounced but shorter-lasting than in the comparable experiment shown in Figure 1H. This discrepancy requires further explanation.

d) A valuable addition would be staining for noradrenergic terminals in the spinal cord for the intersectional approach (Figure 1J), as done in Figures 1F/G. LC projections terminate preferentially in the SDH, whereas A5 projections terminate in the deep dorsal horn (DDH). Staining could clarify whether circuits beyond the LC are being ablated.

e) Furthermore, different LC neurons often mediate opposite physiological outcomes depending on their projection targets—for example, dorsal LC neurons projecting to the prefrontal cortex PFCx are pronociceptive, while ventral LC neurons projecting to the SC are antinociceptive (PMIDs: 29027903, 34344259, 36625030). Given this functional diversity, direct injection into the LC is likely to result in nonspecific effects.

Conclusion on Specificity: The authors are strongly encouraged to address these limitations directly, as they significantly affect the validity of the conclusions regarding the LC pathway. Providing more robust evidence, acknowledging experimental limitations, and incorporating complementary analyses would greatly strengthen the manuscript.

(2) Discrepancies in Data

a) Figures 1B and 1E: The behavioural effect of stress on PWT (Figure 1E) persists for 120 minutes, whereas Ca^{2+} imaging changes (Figure 1B) are only observed in the first 20 minutes, with signal attenuation starting at 30 minutes. This discrepancy requires clarification, as it impacts the proposed mechanism.

b) Figure 4E: The effect is barely visible, and the tissue resembles "Swiss cheese," suggesting poor staining quality. This is insufficient for such an important conclusion. Improved staining and/or complementary staining (e.g., cFOS) are needed. Additionally, no clear difference is observed between Stress+Ab stim. and Stress+Ab stim.+CPT, raising doubts about the robustness of the data.

c) Discrepancy with Existing Evidence: The claim regarding the pronociceptive effect of LC → SDH-NAergic signalling on mechanical hypersensitivity contrasts with findings by Kucharczyk et al. (PMID: 35245374), who reported no facilitation of spinal convergent (wide-dynamic range) neuron responses to tactile mechanical stimuli, but potent inhibition to noxious mechanical von Frey stimulation. This discrepancy suggests alternative mechanisms may be at play and raises the question of why noxious stimuli were not tested.

(3) Sole reliance on Von Frey testing

The exclusive use of von Frey as a behavioural readout for mechanical sensitisation is a significant limitation. This assay is highly variable, and without additional supporting measures, the conclusions lack robustness. Incorporating other behavioural measures, such as the adhesive tape removal test to evaluate tactile discomfort, the needle floor walk corridor to assess sensitivity to uneven or noxious surfaces, or the kinetic weight-bearing test to measure changes in limb loading during movement, could provide complementary insights. Physiological tests, such as the Randall-Selitto test for noxious pressure thresholds or CatWalk gait analysis to evaluate changes in weight distribution and gait dynamics, would further strengthen the findings and allow for a more comprehensive assessment of mechanical sensitisation.

Overall Conclusion

This study addresses an important and complex topic with innovative methods and compelling data. However, the conclusions rely on several assumptions that require more robust evidence. Specificity of the LC pathway, experimental discrepancies, and methodological limitations (e.g., sole reliance on von Frey) must be addressed to substantiate the claims. With these issues resolved, this work could significantly advance our understanding of astrocytic and noradrenergic contributions to pain modulation.

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